
LECTURE NOTES

Lecture 2: Learning and memory

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=- Short-term memory

- (magic number “7” +/-2, today 4-5)
- **Long-term memory**
 - Explicit (what happened when), Implicit (how)
- Retrieval (recognition, recall, reconstruction)
 - retrieving memories with recall or recognising something from the memory.
 - When you remember a specific thing, like a situation, (reconstruction) to remember other things.

1 Behaviourism

- Classical conditioning (strong cues), like Pavlov's dogs
- Operational conditioning (doves playing pingpong)

2 More generally, how do we learn

- **Priming**
 - When two brain processes are activated together or in immediate succession, the activation of one spreads to the other
- **Competition among predictors**
 - Our world is complex – in each moment, myriads of stimuli could be relevant
 - To save energy, our brain thus sets them in direct competition to each other – only the best ones survive
 - If there is already a relevant stimulus, new ones are blocked
 - Only stimuli that provide unique new information are learned
- **Configural learning**
 - Our brain loves efficiency – so we often learn things as a “bundle” (Schema theory)
 - Consider learning how someone looks like – for example Zendaya – you don't learn her hair and eye colour separately. This allows you to recognize her even with her “dune” eyes
- **Generalization and similarity**
 - Projecting our knowledge to new situation
 - example, you know how to bike in sunny weather, so you kinda know how to bike in rain.
 - The lower the similarity between situations or stimuli, the better do we learn to distinguish between them
 - Shift your attention to different dimension
 - Shift your attention to different values on the dimension
- **Note (some exceptions)**
 - **Reinforcement learning**

- seems to be more powerful – Just watching soccer doesn't make you a successful player
- **Interval of stimuli**
 - The interval between the stimuli matters – and variety keeps your brain from zooming out
- **Unlearning**
 - If a stimulus is paired with a response and the response never comes, the pairing goes extinct over time